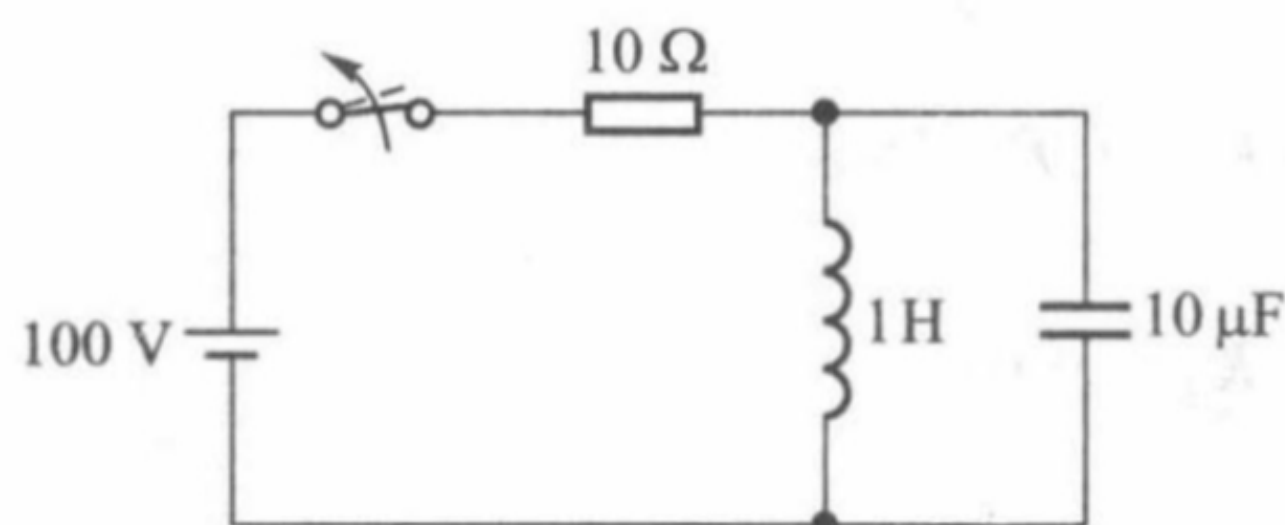


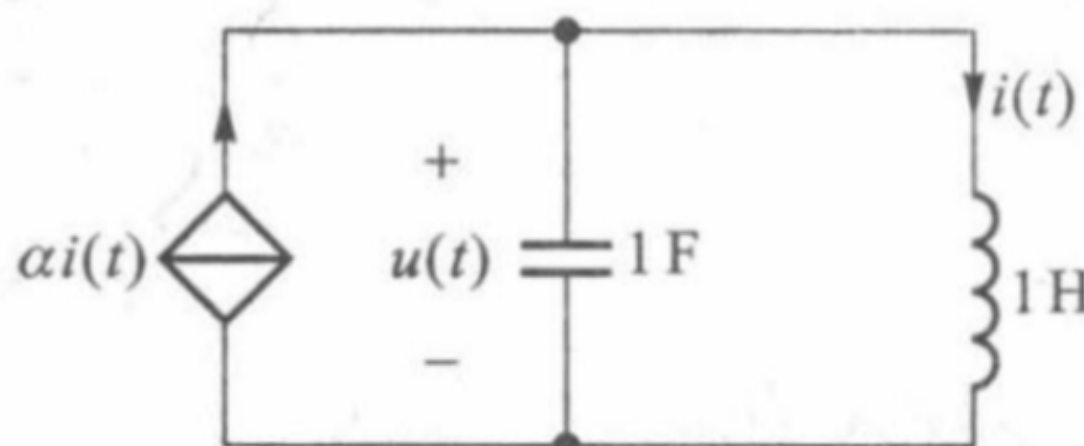
第七章

7-2 电路如图题 7-1 所示, 开关在 $t=0$ 时打开, 打开前电路已处于稳态。求 $i_L(t)$ 、 $u_C(t)$, $t \geq 0$ 。

7-3 电路如图题 7-2 所示, 若 $i(t) = 5\sin(3t)$, $t \geq 0$; $i(t) = 0$, $t < 0$ 。试确定 $i(0)$ 、 $u(0)$ 以及 α 值。



图题 7-1



图题 7-2

$$i_L(0_-) = i_L(0_+) = 10 \text{ A}$$

$$u_C(0_-) = u_C(0_+) = 0 \text{ V}$$

断开后, $R=0$

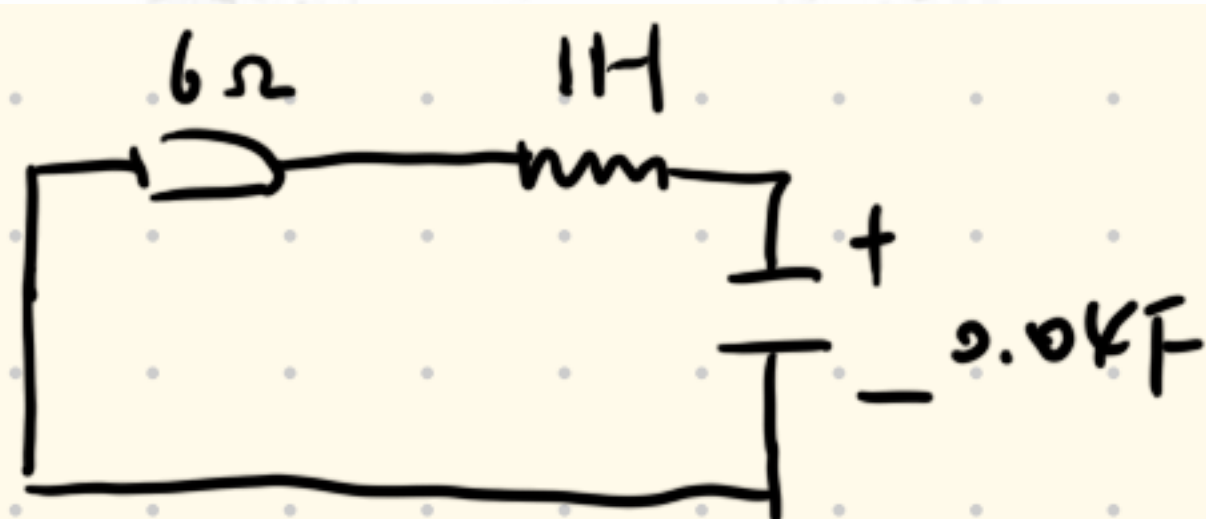
$$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-5}}} \approx 316 \text{ rad/s}$$

$$i_L(t) = i(0) \cdot \cos \omega t = 10 \cos 316t \text{ (A)}$$

$$u_C(t) = -L \cdot \frac{di_L}{dt} = -1 \cdot 316 \cdot 10 \cdot \sin 316t = -3160 \sin 316t \text{ (V)}$$

16-66

7-5 已知 RLC 串联电路中 $R=6 \Omega$, $L=1 \text{ H}$, $C=0.04 \text{ F}$, $i(0)=4 \text{ A}$ 和 $u_C(0)=-4 \text{ V}$ (u_C 与 i 关联参考方向)。试求 $i(t)$ 、 $u_C(t)$, $t \geq 0$ 。



$$RC \cdot \frac{du_C}{dt} + LC \cdot \frac{d^2 u_C}{dt^2} + u_C = 0$$

特征方程 $0.04x^2 + 6 \times 0.04x + 1 = 0$

$$\Rightarrow x^2 + 6x + 25 = 0$$

$$x = \frac{-6 \pm \sqrt{36 - 100}}{2} = \frac{-6 \pm \sqrt{-64}}{2} = -3 \pm 4j$$

$$\therefore u_C(t) = e^{-3t} (A_1 \cos 4t + A_2 \sin 4t)$$

$$\because u_C(0) = -4 \text{ V}$$

$$i(0) = 4 \text{ A}$$

$$\therefore \begin{cases} A_1 \cdot 1 = -4 \\ C \cdot \frac{du_C(t)}{dt} \Big|_{t=0} = 4 \end{cases} \Rightarrow \begin{cases} A_1 = -4 \\ (4A_2 + 12) \times 1 \times 0.04 = 4 \end{cases} \Rightarrow \begin{cases} A_1 = -4 \\ A_2 = 22 \end{cases}$$

$$\frac{du_C(t)}{dt} = e^{-3t} (-3)(A_1 \cos 4t + A_2 \sin 4t)$$

$$+ e^{-3t} (-4A_1 \sin 4t + 4A_2 \cos 4t)$$

$$= (-3A_2 - 4A_1) e^{-3t} \sin 4t + (4A_2 - 3A_1) e^{-3t} \cos 4t$$

$$(RC)^2 - 4LC$$

$$LCR^2 - C - 4L$$

$$2\sqrt{\frac{L}{C}}$$

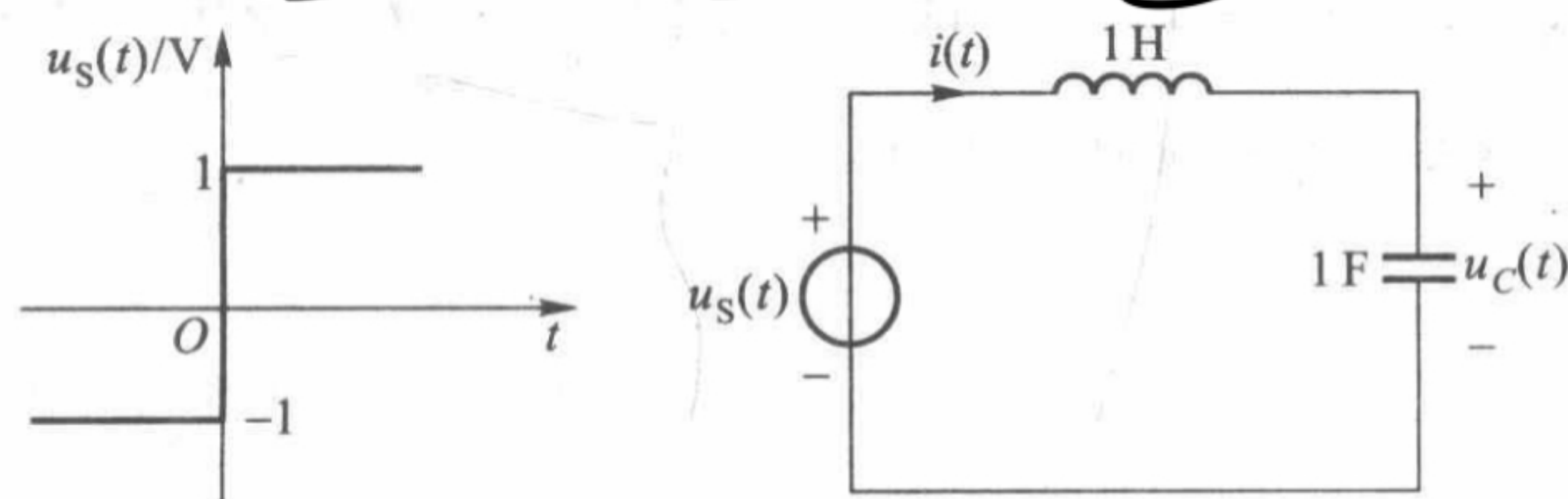
$$\therefore t \geq 0 \text{ ms}$$

$$u_C(t) = e^{-3t} (-4 \cos 4t + 22 \sin 4t) \text{ V}$$

$$i(t) = 0.04 e^{-3t} (-50 \sin 4t + 100 \cos 4t)$$

$$= e^{-3t} (4 \cos 4t - 20 \sin 4t) \text{ A}$$

7-7 电路如图题 7-3 所示, 当 $t < 0$ 时, $u_s(t) = -1 \text{ V}$ 。在 $t = 0$ 时, $u_s(t)$ 突然增至 1 V , 以后即一直保持为此值。试求电路中各元件的电压和电流, 并绘波形图。



图题 7-3

$$i_L(0-) = 0 \text{ A} = i_L(0+)$$

$$u_C(0-) = -1 \text{ V} = u_C(0+)$$

无源元件响应:

$$LC \cdot \frac{d^2 u_C}{dt^2} + u_C = 0$$

$$X = \pm 1j$$

$$\text{齐次解: } u_{Ch}(t) = k_1 \cos t + k_2 \sin t$$

$$\text{特解: } u_{Cp}(t) = 1 \text{ V}$$

$$\therefore u_C(t) = u_{Ch}(t) + u_{Cp}(t) = k_1 \cos t + k_2 \sin t + 1 \quad u_L(t) = 2 \cos t \quad (t \geq 0)$$

$$2: u_C(0) = k_1 + 1 = -1$$

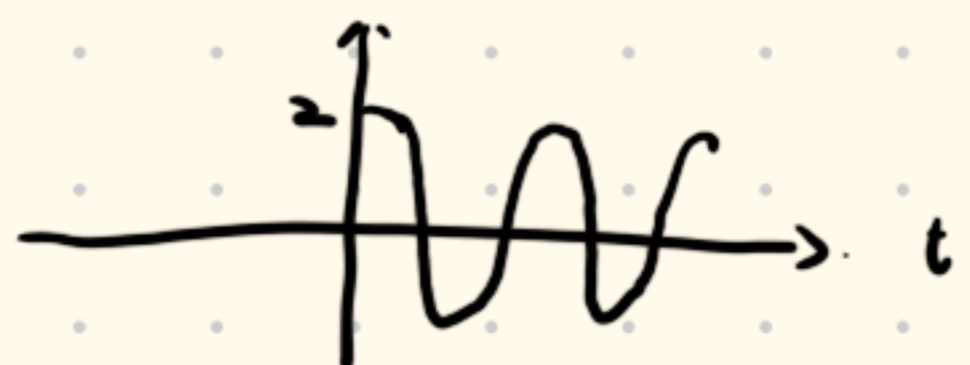
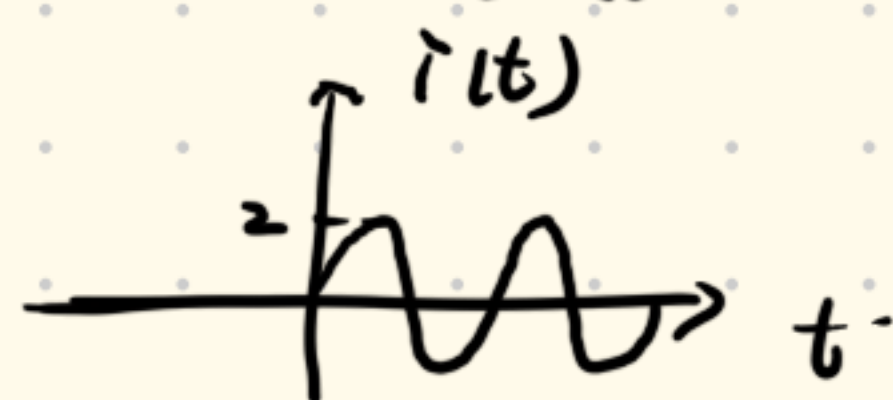
$$i(0) = C \cdot \frac{du_C}{dt} \bigg|_{t=0} = 1 \cdot (k_2) = 0$$

$$\Rightarrow \begin{cases} k_1 = -2 \\ k_2 = 0 \end{cases}$$

$$\therefore u_C(t) = -2 \cos t + 1 \quad (t \geq 0)$$

$$i_L(t) = 2 \sin t \quad (t \geq 0)$$

1 H 电感



1 F 电容

